SANJANA FERNANDO

Software Engineer

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PROFESSIONAL SUMMARY

Computer Engineering undergraduate with a strong foundation in data structures, algorithms, full-stack development, and machine learning. Experienced in building responsive applications and collaborating in agile teams to deliver scalable, secure, user-focused solutions.

TECHNICAL SKILLS

- **Web Development:** MongoDB, Express.js, React.js, Node.js, Next.js, Vite.js, TypeScript, REST APIs, Tailwind CSS, Spline
- **Mobile Development:** Flutter, Dart, Firebase
- **Machine Learning & AI:** Python, Data Preprocessing, Pandas, NumPy, Scikit-learn, PyTorch, Reinforcement Learning, Computer Vision, MediaPipe, OpenCV
- **Security & Tools:** Git, Web Security Models, Cryptography (Encryption/Hashing)

PROFESSIONAL EXPERIENCE

Software Engineering Intern *May 2025 – Feb 2026*
BotCalm (PVT) Ltd. *Sri Lanka*

- Developed high-performance web applications using Vite.js, integrating complex UI animations with Framer Motion to deliver fluid, highly interactive, and responsive front-end user interfaces.
- Co-engineered a full-stack luxury marketplace platform using Next.js, implementing 10+ core front-end UI components and optimizing back-end API logic to reduce page load times by 15%.
- Collaborated in an agile team to build a casino fraud detection project using Next.js, developing full-stack components and secure data flows to identify and flag anomalous user behavior.

Bank Trainee *Dec 2021 – Feb 2022*
Peoples Bank *Kollupitiya Coop House,Sri Lanka*

- Accurately processed 150+ high-volume financial transactions daily as the main cashier, maintaining a 100% accuracy rate in daily cash balancing and compliance auditing.
- Handled critical customer problem-solving scenarios under pressure, maintaining professional communication and strict adherence to banking regulations.

EDUCATION

Bachelor of Science in Computer Engineering Specialization **Expected Graduation: Sep 2026**
University of Ruhuna | Department of Electrical and Information Engineering
Final CGPA: 3.33 / 4.00

- **Relevant Coursework:** Web Development, Project Management, Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Artificial Intelligence & Machine Learning, UI/UX, Networking, Analog & Digital Electronics

GCE Advanced Level (2020)
Physics – A, Physics – A, Chemistry – B *Z-score: 1.9*

KEY PROJECTS

Hospital Bed Allocation & Management Platform using Reinforcement Learning

- Developed a full-stack optimization platform combining a Next.js dashboard with a Python backend to deploy a custom Reinforcement Learning agent, successfully automating hospital bed allocation to maximize asset utilization and reduce patient wait times.
- **Technologies:** Next.js, React.js, Tailwind CSS, Python, PyTorch, DoubleDQN (DDQN), NumPy, Pandas, SciPy

BookTalk – AI Chat Application *GitHub*

- Built a Next.js chat application featuring a clean web UI to interact with a Groq-hosted Llama model, leveraging advanced streaming capabilities.
- **Technologies:** Next.js, React.js, TypeScript, Tailwind CSS, Vercel AI SDK, Groq, OpenAI SDK

AI-Powered Hand Gesture Virtual Mouse *GitHub*

- Engineered an AI-driven computer vision application that processes real-time webcam streams to track hand landmarks and classify gestures for system control.
- **Technologies:** Python, PyTorch, MediaPipe, OpenCV, Deep Learning (Neural Networks)

Secure File Drop *GitHub*

- Co-engineered a secure full-stack web application featuring a robust authentication system and granular access controls to guarantee protected file transfers, implementing proactive defenses against unauthorized access.
- **Technologies:** Cryptography (Encryption/Hashing), Web Security Models, HTML, CSS, Git

E-Chess *GitHub*

- Developed a full-stack web platform designed to manage and participate in competitive, real-time chess tournaments.
- **Technologies:** MongoDB, Express.js, Vite.js, Node.js, Chess.js

Fitness Tracking Mobile Application *GitHub*

- Created a mobile application for comprehensive fitness tracking, providing users with specialized meal plans, exercise tracking, and workout routines.
- **Technologies:** Flutter, Dart, Firebase

Currency Exchange Market *GitHub*

- Designed and built a prototype full-stack platform for monitoring and simulating currency exchange transactions.
- **Technologies:** MongoDB, Express.js, React.js, Node.js, Tailwind CSS

CERTIFICATIONS

- Python for Data Science and Machine Learning Bootcamp – Udemy
- The Ultimate React Course: React, Next.js, Redux & More – Udemy
- Flutter & Dart - The Complete Guide – Udemy
- React, NodeJS, Express & MongoDB - The MERN Fullstack Guide – Udemy

EXTRA-CURRICULAR ACTIVITIES

- **IEEEXtreme 18.0 Hackathon (Logistics Lead):** Orchestrated operations and resource allocation for the global programming hackathon, helping the university secure its highest-ever ranking.
- **EMINENCE 4.0 (Logistics Committee Member):** Streamlined logistical workflows and event deployment, directly contributing to winning the Best Affinity Group Project Award in the IEEE Sri Lanka Section.
- **Project "Idora Yathra" (Co-Manager):** Co-managed a community outreach initiative that donated over 500 books, conducted youth leadership workshops, and coordinated local student meals.
- **Soft Skill Development Course:** Completed an intensive professional development program focused on leadership, collaborative problem-solving, and workplace communication.

REFERENCES

Dr. Geeth Priyankara
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